Matias C.Y. Relyea

CONTACT INFORMATION
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EDUCATION

University of North Carolina at Chapel Hill, Chapel Hill, North Carolina

B.S. Mathematics; Minor in violin performance (expected graduation, May 2029).

Relevant Coursework: Undergraduate: Differential Geometry; Topology (Duke University); Algebraic Geometry; Algebraic Structures; Advanced Calculus. Graduate: Modules, Linear Algebra, and Groups; Groups, Representations, and Fields; Differentiable Manifolds.

PUBLICATIONS

[1] Matias C.Y. Relyea. The Quadratic and Cubic Characters of 2. ***Mathematics Magazine***. Published online January 2026. [arXiv: 2410.21646] [DOI: 10.1080/0025570X.2025.2574223].

PREPRINTS

[2] Harin Jung, Aiden Novick, Matias C.Y. Relyea. Improved Poly-Log Diameter Bounds for $\mathrm{SL}_2(\mathbb{Z}_p)$ via Higher Commutators. *In preparation*.

[3] Matias C.Y. Relyea. On Finite Fields and Higher Reciprocity (2024). [arXiv: 2407.03559].

PRESENTATIONS

[4] Harin Jung, Aiden Novick, Matias C.Y. Relyea. "A New Upper Bound for the Polylog Exponent of the Diameter of $\mathrm{SL}_2(\mathbb{Z}_p)$ via Higher Commutators." Contributed Paper Session on Algebraic Geometry, Joint Mathematics Meetings (JMM), Chicago, Illinois, January 2026 (*accepted*).

[5] Harin Jung, Aiden Novick, Matias C.Y. Relyea. "Improved Poly-Log Diameter Bounds for $\mathrm{SL}_2(\mathbb{Z}_p)$ via Higher Commutators." Pi Mu Epsilon (PME) Undergraduate Poster Session, Joint Mathematics Meetings (JMM), Chicago, Illinois, January 2026 (*accepted*).

[6] Matias C.Y. Relyea. "Reciprocity Laws" (*chalkboard lecture series*). Carolina Math Club, Chapel Hill, North Carolina. March and April 2026.

[7] Matias C.Y. Relyea. "On Finite Fields and Higher Reciprocity" (*poster presentation*). NCSSM Research Symposium Spring 2024.

RESEARCH EXPERIENCE

Einstein Institute of Mathematics, Hebrew University of Jerusalem (HUJI)

REU Visiting Research Fellow, Summer 2026

- Extended the Solovay-Kitaev theorem for compact Lie groups, proving new diameter bounds for $SL_2(\mathbb{Z}/p^n\mathbb{Z})$ using Elkasapy words and higher commutators. Quantum computation, Lie theory, group theory, and algebraic number theory. Advisor: Shai Evra. *Manuscript in preparation.*

University of North Carolina at Chapel Hill, Chapel Hill, North Carolina

Undergraduate Researcher, Fall 2026—Present

- Ongoing research in birational geometry; investigating Weyl group actions on the blowups of n-dimensional projective space. Advisor: Olivia Dumitrescu.

Directed Reading Program, Fall 2025 and 2026

- Algebraic Topology, Homology and Cohomology, Fall 2026
- Category Theory, Fall 2025

North Carolina School of Science and Mathematics (NCSSM), Durham, North Carolina

Independent studies on number theory and algebraic number theory, producing one arXiv preprint and one peer-reviewed publication. August 2023—May 2025. Mentored by Professor Tamar Avineri.

Euler Circle, Mountain View, California

Attended Summer 2022 Independent Research and Paper Writing course. Independently produced 44-page expository paper on quadratic reciprocity. Mentored by Professor Simon Rubinstein-Salzedo.

Seminars and Conferences Attended

[8] 25th Midrasha Mathematicae: Groups, Expanders, and Codes. Israel Institute for Advanced Studies (IIAS), 14-18 June 2026.

Other Writing

[9] Peyton Jackson, Tatiana Medved, and Matias C.Y. Relyea. "Can You Even?" An Exploration of Polyominos. ***NCSSM Broad Street Scientific, 2024-2025*** (link). Published online May 2026. [PDF].

[10] Matias C.Y. Relyea. Proofs and Applications of Quadratic Reciprocity. Summer 2022, ***Euler Circle***. [PDF].

Teaching and Service

Mathematics Institute of the Triangle, Chapel Hill, North Carolina

Mathematics Tutor and Teaching Assistant, Fall 2025—Present

- Provided one-on-one tutoring to students located in the Chapel Hill area. Assisted with grading and led short problem-solving workshops.

University of North Carolina at Chapel Hill, Chapel Hill, North Carolina

Undergraduate Learning Assistant

- MATH 381 (Discrete Mathematics), Fall 2025.

Chapel Hill Math Circle, Chapel Hill, North Carolina

Program Volunteer, Fall 2025—Present

- Collaborated with UNC graduate students to assist and lead teaching for various levels of mathematics circle coursework.

Technological Skills

Proficient in LaTeX, familiar with Python.